

E-scooter Battery Swap Model (Macro)

Model Description:

- 1- Model represents a free-floating e-scooter renting operation. It considers the battery swapping and relocation of e-scooter as possible service operation. It takes starting status as input.
- 2- Service vehicles are used to replace the batteries and/or relocate the e-scooter to better fulfill the known demand.
- 3- Penalty on unfulfilled demand during the relocation operation.
- 4- Before the start of the decision-making period, e-scooters with sufficient or insufficient battery level for each region are determined.
- 5- System is not expected to fulfill all the trip request.
- 6- Trip requests are characterized by their start point, end point, start time, finish time, and energy requirement (battery percentage). However, for this model demand is simplified as a number of e-scooters required in a region.
- 7- Processing time for battery replacement and e-scooter relocation (collection or drop-off) for each region are estimated a-priori. Several heuristics can be used to estimate this value.
- 8- Service vehicles have fixed carrying capacity for the e-scooters and batteries, which is not interchangeable.

Mathematical model:

Sets and indices

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| R | Set of regions, indexed by r . |
| N | Set of nodes representing location of hub, and region-centers $N = \{0\} \cup R$ |
| V | Set of service vehicles, indexed by v . |

Parameters

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|---------------|--|
| D_r | Expected demand of region r . |
| $\check{S}_r$ | Number of non-functional scooters (battery below level 2) region r . |
| $\hat{S}_r$ | Number of functional scooters (battery equal or above level 2) region r . |
| A | Number of extra batteries in the system ($A < S $). |
| C_v^b | Battery carrying capacity (number of batteries) of vehicle v . |
| C_v^s | Electric scooter carrying capacity (number of electric scooter) of vehicle v . |
| M | A sufficiently large positive number. |

t_{ij}	Total time (minutes) needed to travel from node i to node j .
T_r	Processing time (minutes) for region r .
λ	Travel cost (dollar/minute).
λ'	Penalty for unmet demand.
λ''	Penalty associated with delay in e-scooter readiness ($\lambda'' = \lambda' / \text{period length}$)

Decision variables

x_r	Number of e-scooter selected for the battery swap in region r .
p_{rl}	Number of e-scooter to be moved from region r to region l .
y_{rl}	$\begin{cases} 1, & \text{if atleast one e – scooter is to be moved from region } r \text{ to region } l. \\ 0, & \text{otherwise.} \end{cases}$
u_v	$\begin{cases} 1, & \text{if vehicle } v \text{ is required for service operations.} \\ 0, & \text{otherwise.} \end{cases}$
z_{rv}	$\begin{cases} 1, & \text{if region } r \text{ is assigned to vehicle } v \text{ for service operations.} \\ 0, & \text{otherwise.} \end{cases}$
w_{vij}	$\begin{cases} 1, & \text{if vehicle } v \text{ travels from node } i \text{ to node } j \text{ during service operation.} \\ 0, & \text{otherwise.} \end{cases}$
q_{vi}	Number of Electric scooters in vehicle v after visiting node i .
m_{vi}	Departure time for vehicle v from node i .
d_r	Unmet demand of region r .
h_r	Penalty factor for region r .

Objective function:

$$(MIN) Z = \sum_{v \in V} \sum_{i \in N} \sum_{j \in N} \lambda * w_{vij} + (\lambda' * d_r) + \sum_{r \in R} \sum_{v \in V} (\lambda'' * h_r * m_{vr}) \quad (1)$$

Subject to,

$$x_r \leq \tilde{S}_r \quad \forall r \in R \quad (2)$$

$$\sum_{r \in R} x_r \leq A \quad \forall r \in R \quad (3)$$

$$\sum_{l \in R} p_{rl} \leq x_r + \hat{S}_r \quad \forall r \in R \quad (4)$$

$$\sum_{v \in V} z_{rv} \leq 1 \quad \forall r \in R \quad (5)$$

Objective function minimizes the travel cost, penalty associated with lost demand and delay in readiness (Eq. 1). **Based on the necessity**, Constraint-2 limits the number of e-scooter that are selected for battery swap for a single region. Constraint-3 ensures that the number of e-scooter selected for battery swap are below the available fully-charged batteries. Similarly, for each region, Constraint-4 ensures that number of the e-scooters selected for relocation are below the number of sufficiently charged scooter. Constraint-5 ensures that each region is assigned exactly one service vehicle for the necessary service operations. *In contrast, not applicable here, the Micro-model does not restrict more than one vehicle to fulfill the demand of a particular region.*

$$\sum_{r \in R} z_{rv} \leq M * u_v \quad \forall v \in V \quad (6)$$

$$\sum_{i \in R} w_{vi\{0\}} = u_v \quad \forall v \in V \quad (7)$$

$$\sum_{j \in R} w_{v\{0\}j} = u_v \quad \forall v \in V \quad (8)$$

$$\sum_{j \in N} w_{vij} = z_{iv} \quad \forall v \in V, i \in R \quad (9)$$

$$\sum_{i \in N} w_{vij} = z_{jv} \quad \forall v \in V, j \in R \quad (10)$$

$$m_{vi} + t_{ij} + T_r - m_{vj} \leq M * (1 - w_{vij}) \quad \forall v \in V, i \in N, j \in R \quad (11)$$

Constraint-6 identifies the vehicles assigned to at least one region for the service operations and makes them available for routing. Constraints 7 and 8 ensure that each vehicle starts and ends its route at the hub. Similarly, Constraint 9 and 10 ensure that vehicles have a continuous route. Constraint 11 sub-tour elimination constraint and calculates the departure time for each node on the vehicle route.

$$p_{rl} \leq M * y_{rl} \quad \forall r \in R, l \in R \quad (12)$$

$$m_{vr} - m_{vl} \leq M * (1 - y_{rl}) \quad \forall v \in V, r \in R, l \in R \quad (13)$$

$$2 - z_{rv} - z_{lv} \leq M * (1 - y_{rl}) \quad \forall v \in V, r \in R, l \in R \quad (14)$$

Constraints 12 to 14 are used to ensure the alignment of e-scooter relocation to the movement of service vehicles. Based on the relocation decision, Constraint 12 establishes the dependency between two regions. Then Constraint 13 is used to assigned same service vehicle to these dependent regions. Similarly, Constraint 14 ensures the departure time for the dependent regions are in order.

$$q_{v0} = 0 \quad \forall v \in V \quad (15)$$

$$q_{vi} + \sum_{l \in R} p_{jl} - \sum_{r \in R} p_{rj} - q_{vj} \leq M * (1 - w_{vij}) \quad \forall v \in V, i \in N, j \in R \quad (16)$$

$$q_{vi} \leq C_v^s \quad \forall v \in V, i \in N \quad (17)$$

$$\sum_{r \in R} (z_{rv} * x_r) \leq C_v^b \quad \forall v \in V \quad (18)$$

$$x_r + \sum_{l \in R} p_{lr} - \sum_{l \in R} p_{rl} \leq h_r \quad \forall r \in R \quad (19)$$

$$\hat{S}_r + x_r + \sum_{l \in R} p_{lr} - \sum_{l \in R} p_{rl} + d_r - d_r = D_r \quad \forall r \in R \quad (20)$$

Constraints 15 and 16 are used to find the number of e-scooter on the service vehicle during its route. Constraint 17 restrict this quantity to be below the carrying capacity of the vehicle. Constraint 18 (non-linear) does the same with regards to the battery carrying capacity. Finally, Constraints 19 and 20 are used to calculate the penalties. Where, Constraint 19 finds the number of e-scooters with delayed activation, Constraint 20 calculates the outright lost demand.